

EXPERIMENT 10

AIM:

To understand and perform basic array functions using the NumPy library in Python.

CODE:

```
import numpy as np
```

```
# Create 1-D array using values from the dataset
```

```
array = np.array([15, 15, 16, 16, 17, 17, 18, 18, 19])
```

```
print("1. Original Array:")
```

```
print(array)
```

```
# Square root
```

```
print("\n2. Square Root of Array:")
```

```
print(np.sqrt(array))
```

```
# Number of dimensions
```

```
print("\n3. Number of Dimensions:")
```

```
print(array.ndim)
```

```
# Reshape into 3x3
```

```
new_array = array.reshape(3, 3)
```

```
print("\n4. Reshaped Array:")
```

```
print(new_array)
```

```
# Number of dimensions after reshape
```

```
print("\n5. Number of Dimensions after Reshape:")
```

```
print(new_array.ndim)
```

```
# Ravel
```

```
print("\n6. Ravelled Array:")
```

```
print(new_array.ravel())
```

```
# Reshape again
```

```
newm = new_array.reshape(3, 3)
```

```
print("\n7. New Array:")
```

```
print(newm)
```

```
# Slicing operations
```

```
print("\n8. newm[2,1:3]:")
```

```
print(newm[2,1:3])
```

```
print("\n9. newm[1:2,1:3]:")
```

```
print(newm[1:2,1:3])
```

```
print("\n10. new_array[0:3,0:0]:")
```

```
print(new_array[0:3,0:0])
```

```
print("\n11. new_array[0:2,0:1]:")
```

```
print(new_array[0:2,0:1])
```

```
print("\n12. new_array[0:3,0:1]:")
```

```
print(new_array[0:3,0:1])
```

```
print("\n13. new_array[1:3]:")
```

```
print(new_array[1:3])
```

OUTPUT:

1. Original Array:

```
[15 15 16 16 17 17 18 18 19]
```

2. Square Root of Array:

```
[3.87298335 3.87298335 4.      4.      4.12310563 4.12310563  
4.24264069 4.24264069 4.35889894]
```

3. Number of Dimensions:

```
1
```

4. Reshaped Array:

```
[[15 15 16]
```

[16 17 17]

[18 18 19]]

5. Number of Dimensions after Reshape:

2

6. Ravelled Array:

[15 15 16 16 17 17 18 18 19]

7. New Array:

[[15 15 16]

[16 17 17]

[18 18 19]]

8. newm[2,1:3]:

[18 19]

9. newm[1:2,1:3]:

[[17 17]]

10. new_array[0:3,0:0]:

[]

11. `new_array[0:2,0:1]:`

`[[15]`

`[16]]`

12. `new_array[0:3,0:1]:`

`[[15]`

`[16]`

`[18]]`

13. `new_array[1:3]:`

`[[16 17 17]`

`[18 18 19]]`

RESULT:

The NumPy array functions were successfully performed, and the array dimensions, reshaped arrays, and slicing operations were obtained.